

# Aditya Saini

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## EDUCATION

### Purdue University

West Lafayette, IN

*B.S. Computer Science (Systems Programming)*

*Graduation: May 2028 (Expected)*

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

## EXPERIENCE

### Robotic Safety Research Assistant

January 2026 – Present

*PurSec, Purdue Computer Science*

*West Lafayette, IN*

- Researched jailbreak vulnerabilities in MLLM-controlled home-assistant robots, uncovering multi-step physics attacks that chain individually-safe actions into unsafe outcomes to defeat existing safety systems
- Engineered 10+ AI2-THOR multi-step physics attacks (chain-reaction, fire, and flooding hazards) that bypass state-of-the-art guardrails while every intermediate step passes inspection
- Built a dual-brain safety baseline pairing RoboGuard (GPT-4o rules to Linear Temporal Logic compiled into Spot Büchi automata) with ASIMOV (GPT-4o Vision over frames + a Robot Constitution) to red-team its blind spots
- Demonstrated attacks defeating both brains and ran reproducible evaluations via a Dockerized headless (Xvfb/FFmpeg) pipeline on AWS EC2 to quantify per-step and end-to-end failure rates

### Embedded Engineer

Sep. 2025 – Present

*Purdue Electric Racing*

*West Lafayette, IN*

- Developed modular RTOS-based vehicle firmware in C, integrating drivers and performing in-circuit debugging
- Implemented runtime telemetry and data acquisition to validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms, standardizing high-frequency sensor communication
- Developed Nextion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

### Advanced Mathematics Teacher

June 2023 – July 2025

*Kumon North America*

*Edison, NJ*

- Developed tailored pedagogical strategies to explain complex mathematical concepts to a diverse student body, ensuring mastery of core principles for over 20+ concurrent students.
- Managed high-volume academic data entry and performance tracking, utilizing center-specific software to generate progress reports and identify learning bottlenecks for student development.

## PROJECTS

### Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

### FRC Team 2554 | Java, OpenCV, Jetson Orin Nano, CUDA

September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Developed and led our team's first instance of Swerve Drive, tuned PID/FF loops, and developed auto commands to increase accuracy by 66% using custom vision pipelines to detect AprilTags for robot positioning.
- Managed finances and administrative requirements for the team in conjunction with robot design requirements.

### Tava | React Native, Expo, TypeScript, Firebase, Mapbox, OpenAI

September 2025

- Built a cross-platform social app connecting student builders via an AI-powered search and Tinder-style swipe matching, with a node-edge network graph and interactive Mapbox map backed by Firebase auth and data.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, SQL, JavaScript, HTML/CSS, R

**Embedded:** RTOS, STM32, ESP32, SPI, I2C, UART, CAN, TCP/IP, Arduino, Linux

**Developer Tools:** Git, Docker, Amazon Web Services, Claude Code, Cursor, Agent Orchestration

**Libraries:** pandas, NumPy, Matplotlib, React, Node.js, PyTorch, CUDA, OpenCV